



Sanskar Gupta

Roll No.: 20253248

B.Tech

Computer Science and Engineering

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EDUCATION

- **Motilal Nehru National Institute of Technology Allahabad, Prayagraj** 2025 – 2029
B.Tech, Computer Science and Engineering CGPA: 8.98
- **Modern Public School, Jhansi, U.P** 2024 – 2025
Class XII, CBSE Percentage: 93%
- **Sun International School, Jhansi, U.P** 2022 – 2023
Class X, CBSE Percentage: 94.8%

PERSONAL PROJECTS

- **KnightCode — Full-Stack Competitive Coding Platform** January 2026 – Present
React, Vite, Express.js, Node.js, MongoDB, Docker, Three.js, JWT, Framer Motion GitHub
 - Built a full-stack competitive programming judge from scratch with a custom **Docker-based code execution engine** (no Judge0), supporting C++, Python, and JavaScript with isolated sandboxes (`-net=none`, `-cpus=0.5`, `-memory=256m`).
 - Implemented **parallel test case execution** (batch size of 10 Docker containers) with dynamic per-problem time and memory limits, cutting evaluation time significantly vs. sequential execution.
 - Designed a **3D animated landing page** using Three.js and React Three Fiber with scroll-linked parallax and sacred geometry animations; integrated Monaco Editor (VS Code engine) with a custom “Ancient Codex” theme.
 - Built **JWT-based authentication** with bcrypt password hashing, role-based access control (RBAC), and an admin problem editor with full CRUD.
- **YT RAG Assistant — Local AI Video Q&A** June 2026
Python, FastAPI, React, Ollama, ChromaDB, LangChain GitHub
 - Built a **full-stack local RAG application** that ingests YouTube video transcripts and enables semantic Q&A using fully open-source models via **Ollama** for both embeddings and LLM inference — zero API cost.
 - Designed a **FastAPI** backend pipeline: transcript chunking, vector embedding via Ollama, similarity retrieval from **ChromaDB**, and context-aware response generation with a locally hosted LLM (Llama 3 / Mistral).
 - Developed a **React** frontend with a chat-style interface supporting real-time streamed responses, YouTube URL input, and session-based conversation history.
- **Portfolio Website** March 2026
Next.js, JavaScript, CSS3, Next.js API Routes, Three.js GitHub
 - Developed a personal portfolio using **Next.js 14 (App Router)** with a component-based architecture covering Hero, About, Projects, Experience, and Contact sections.
 - Built a **live coding stats API** (`/api/profiles`) that fetches real-time data from LeetCode (GraphQL) and Codeforces REST APIs, with **1-hour server-side caching** and graceful stale-cache fallback.
 - Integrated an **interactive 3D model** with scroll-linked parallax wrapper and scroll progress indicator for an immersive UX.
- **Quizzon — Android Quiz App** December 2025
Kotlin, XML, Android Studio GitHub
 - Built a native Android quiz application featuring Home, Level Selection, Quiz, and Result pages with real-time scoring and clean, modern UI.
 - Developed as part of Android development learning; APK available for direct download via Google Drive.

TECHNICAL SKILLS

Languages: C++, C, Python, JavaScript, Kotlin, SQL

Frontend: React.js, Next.js, Vite, Three.js, Framer Motion, HTML, CSS

Backend & Databases: Node.js, Express.js, FastAPI, MongoDB, ChromaDB, Mongoose, REST APIs, JWT, bcrypt

AI / ML: LangChain, Ollama, RAG Pipelines, Vector Embeddings

DevOps & Tools: Docker, Git, GitHub, Android Studio, VS Code, Linux

Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems

Areas of Interest: Systems Programming, Competitive Programming, Machine Learning / AI, Web Development

ACHIEVEMENTS & COMPETITIVE PROGRAMMING

- Codeforces** — Rated **Pupil** on Codeforces (handle: Sanskar__G) *2026*
- CodeChef** — Achieved **2-Star** rating (1400+) on CodeChef (handle: sanskar_d) *January 2026*
- LeetCode** — Achieved **Knight** badge with **500+** **Data Structures & Algorithms** problems solved *2026*
- Robomania** — **Trail Blitz** — Secured **3rd Position** in Line Follower Bot competition at MNNIT *2025*